MIL-P-13541(Ord)
13 July 1954
SUPERSEDING
FCP No. 934-53
2 April 1953

MILITARY SPECIFICATION

PACKAGING OF TESTER, DYNAMIC, M12

1. SCOPE

1.1 This specification covers the preservation, packaging, packing, and marking, for Tester, Dynamic, M12 for overseas shipment or domestic shipment for extended storage, domestic shipment for limited storage, and domestic shipment for immediate use.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard, and drawing of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

FF-N-105 - Nails, Wire; and Staples.
FF-S-111 - Screws, Wood.
NN-P-515 - Plywood, Container Grade.
QQ-S-781 - Strapping, Flat; Steel.
UU-C-843 - Cushioning Materials, Cellulosic.
UU-T-111 - Tape; Paper, Gummed (Sealing and Securing).
UU-T-116 - Tape, Paper, Gummed, Water-Resistant.

Military

JAN-P-100 - Packaging and Packing for Overseas Shipment - General Specification.
MIL-A-101 - Adhesive, Water-Resistant, For sealing Fiberboard Boxes.
JAN-P-106 - Boxes, Wood, Nailed (Overseas Type).
JAN-P-108 - Boxes, Fiberboard, Corrugated and Solid (Military Type).
JAN-P-115 - Packaging and Packing for Overseas Shipment - Compound, Sealing, Dip-Coating.
MIL-P-116 - Preservation, Methods of.
JAN-P-117 - Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
JAN-B-121 - Barrier-Materials, Greaseproof.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-127 - Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
MIL-B-131 - Barrier-Material; Water-Vaporproof, Flexible.
JAN-P-133 - Packaging and Packing for Overseas Shipment - Boxes, Set-Up, Paperboard

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MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.
MIL-D-3464 - Desiccant (Activated) in Bags; For Static Dehumidification and Packaging.
MIL-C-5092 - Cement, Rubber (Synthetic and Reclaimed Rubber Base).
MIL-R-6855 - Rubber, Synthetic, Sheet, Molded and Extruded, For Aircraft Application.
MIL-E-10687 - Enamel, Lustreless, Quick-Drying.
MIL-C-17435 - Cushioning Material, Fibrous, Glass.

STANDARD
Military
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING
Army-Navy Aeronautical

AN 7513 - Humidity Indicator, Small.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Oversea shipment and domestic shipment involving extended storage.--

3.1.1 Packaging.--

3.1.1.1 Cable assemblies.--

3.1.1.1.1 The ends of the 2 conductor cable assemblies which do not have a sealed plug or sealed receptacle shall be covered with greaseproof barrier material conforming to type I or II, grade C, class 2 of Specification JAN-B-121. The wrapped ends shall be sealed by dipcoating with a compound conforming to Specification JAN-P-115. After the compound has set, the dip-coated ends shall be overwrapped with 30 pound kraft paper and the overwraps shall be secured with paper tape conforming to Specification UU-T-111.

3.1.1.1.2 One 3, one 37, and the four 2-conductor cable assemblies shall be coiled and immobilized in one fiberboard box, and one 3, one 37, and the 20 conductor cable assemblies shall be coiled and immobilized in another fiberboard box. The size of each fiberboard box shall be 15-3/8 x 15-3/8 x 14-3/4 inches and it shall conform to compliance symbo V1s or V2s, any style except FTC of Specification JAN-P-108. The fiberboard boxes shall be closed by covering all reas of contact between the flaps with water-resistant adhesive

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conforming to Specification MIL-A-101, and by covering all seams and the manufacturer's joint with tape conforming to Specification UU-T-116 or JAN-P-127.

3.1.1.2 Lamps and fuzes.—Each lamp and each fuze shall be individually wrapped with cellulosic cushioning material conforming to any type, any class of Specification UU-C-843. Each group of four lamps, two fuzes (for milliameter), and eight fuzes (for motor and transmitters) shall be placed in a separate waterproof bag conforming to any type, grade A, class b or c of Specification JAN-P-117. The bags shall be closed by heat sealing after the excess entrapped air is removed. The bags shall be immobilized in a set-up paperboard box conforming to any type, any class, any style of Specification JAN-P-133. The box shall be closed by covering the seams with gummed tape conforming to any class of Specification UU-T-111.

3.1.1.3 Dynamic tester.—The dynamic tester shall be packaged in accordance with method IIa of Specification MIL-P-116. The procedure to be followed, which is illustrated in figures 1 and 2, shall be as specified herein.

3.1.1.3.1 A plywood base shall be constructed in accordance with figure 3. Four step bolts (1/2 inch dia. x 1-1/2 inches long) shall be placed through the holes in the base corresponding to the mounting holes on the tester. Four 2 x 2 x 1/32 inch gaskets conforming to class 2, grade 60 of Specification MIL-R-6855 shall be secured to the base over the bolts with rubber cement. The top of the gaskets shall also be covered with rubber cement. The cement shall conform to type 3 of Specification MIL-C-5092.

3.1.1.3.2 A water-vaporproof barrier furnished as flat sheet, with bolt holes provided, and conforming to class 1 of Specification MIL-B-131 shall be placed over the bolts. A set of four gaskets similar to those specified in 3.1.1.3.1 shall be placed over the bolts and secured with rubber cement to the barrier.

3.1.1.3.3 The dynamic tester shall be mounted on the base. A lock washer shall be used under each nut. The exposed threads above the nuts shall be coated with unthinned paint to prevent the nuts from backing off.

3.1.1.3.4 The box containing the lamps and fuzes, a humidity indicator conforming to Drawing AN 7513, and the required units of desiccant conforming to class 2 of Specification MIL-D-3464 shall be secured to the dynamic tester with pressure-sensitive tape conforming to any type, any grade of Specification JAN-P-127.

3.1.1.3.5 The top, sides, and ends of the dynamic tester shall be covered with a sheet of cellulosic cushioning material 3/8 inch thick and conforming to any type, any grade of Specification UU-C-843. The cushioning material shall be secured with pressure-sensitive tape conforming to any type, any grade of Specification JAN-P-127.

3.1.1.3.6 The water-vaporproof barrier shall be closed by heat sealing after the excess entrapped air is removed. The flaps of the water-vaporproof bag shall be secured with pressure-sensitive tape conforming to any type, any grade of Specification JAN-P-127.

3.1.2 Packing.—

3.1.2.1 A reusable shipping container base shall be constructed in accordance with figure 4. Four step bolts (1/2 inch x 9-1/2 inches long) shall be placed through the holes in the base.

3.1.2.2 The plywood base containing the packaged dynamic tester shall be cushioned as illustrated in figure 5. The procedure to be followed shall be as specified herein.

3.1.2.2.1 The support cushioning pad constructed in accordance with figure 6 shall be placed over the bolts and shall rest on the shipping container base.

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3.1.2.2.2 The packaged tester (with attached lamps and fuzes) shall be placed over the bolts and shall rest on the support cushioning pad.

3.1.2.2.3 The two hold down cushioning pads constructed in accordance with figure 7 shall be placed over the bolts and shall rest on the base of the dynamic tester.

3.1.2.2.4 The two plywood hold down blocks constructed in accordance with figure 8 shall be placed over the bolts and shall rest on the hold down cushioning pads.

3.1.2.2.5 A flat washer, lock washer, and nut shall be placed over each bolt. The nuts shall be tightened until the height from the shipping container base to the top surfaces of the hold down blocks is 6-1/4 inches. The exposed threads above the nuts shall be coated with unthinned paint to prevent the nuts from backing off.

3.1.2.3 The ends, sides, and top of the shipping container, and the plywood separator for the shipping container shall be constructed in accordance with figures 9, 10, 11, and 12 respectively.

3.1.2.4 As illustrated in figure 13, the ends and sides of the shipping container shall be nailed to the base and nailed together, and the plywood separator shall be inserted in the ends.

3.1.2.5 The packaged cable assemblies shall be immobilized in the shipping container; and, as illustrated in figure 13, the top of the shipping container shall be secured with screws and the container shall be strapped in accordance with figure 19.

3.1.2.6 The reusable container shall be covered with one coat of enamel, conforming to Specification MIL-E-10687.

3.2 Domestic shipment for limited storage.--

3.2.1 Packaging.--Packaging shall be as specified in 3.1.1 except that the dipcoating specified in 3.1.1.1.1 shall be omitted.

3.2.2 Packing.--Packing shall be as specified in 3.1.2.

3.3 Domestic shipment for immediate use.--

3.3.1 Packaging.--

3.3.1.1 Cable assemblies.--Cable assemblies shall be packaged in accordance with 3.1.1.1 except that dipcoating specified in 3.1.1.1.1 shall be omitted.

3.3.1.2 Lamps and fuzes.--The lamps and fuzes shall be packaged as specified in 3.1.1.2.

3.3.1.3 Dynamic tester.--

3.3.1.3.1 A plywood base shall be constructed in accordance with figure 3. Four step bolts (1/2 inch dia. x 1-1/2 inches long) shall be placed through the holes in the base corresponding to the mounting holes on the tester. The dynamic tester shall be mounted on the base. A lock washer shall be used under each nut.

3.3.1.3.2 The box containing the lamps and fuzes shall be secured to the dynamic tester with pressure-sensitive tape conforming to any type, any grade of Specification JAN-P-127.

3.3.1.3.3 A shroud fabricated from waterproof barrier material conforming to type E-2 of Specification JAN-P-125 shall be placed over the tester. The seams of the shroud shall be sealed with water-resistant adhesive conforming to Specification MIL-A-140, and the seams and flaps shall be covered with pressure-sensitive tape. The shroud shall be secured to the bottom of the tester with pressure-sensitive tape. The tape shall conform to type II or III, any grade of Spedification JAN-P-127.

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3.3.2 Packing - Packing shall be as specified in 3.1.2.

3.4 Marking

3.4.1 Marking shall be in accordance with Standard MIL-STD-129.

3.4.2 The interior packages (bags, paperboard box, and fiberboard boxes) shall be identified by the quantities, nomenclatures, and stock numbers of their contents. The cable assemblies within the fiberboard boxes shall be similarly identified using paper or cloth tags.

3.4.3 The serial number of the tester shall be included in the permanent marking below the nomenclature and shall be indicated as follows: "SERIAL NO 65".

3.4.4 Special handling markings and preservation markings (when applicable) shall be placed on the shipping container as shown in figure 14.

3.4.5 Special marking - Shipments for DOMESTIC FOR IMMEDIATE USE or DOMESTIC FOR LIMITED STORAGE shall be plainly marked as such on the shipping container.

3.5 Sample pack - A preproduction sample pack shall be made in accordance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling - The sample pack shall be inspected and approved by a Government inspector prior to packaging and packing the items for shipment. Production packs shall be tested at random where the Government inspector considers it necessary to conduct tests to assure conformance with the requirements of this specification.

4.2 Inspection - All methods and materials shall be subject to inspection by a Government inspector.

4.3 Test procedures - At the discretion of the Government inspector, the preproduction sample pack and any production packs selected for test, shall be tested in accordance with the tests specified in Specification MIL-P-116 and JAN-P-100.

5. PREPARATION FOR DELIVERY

5.1 (Not applicable.)

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number, and date of this specification, and the type of shipment to be made.

6.2 Identification data

	Nomenclature	Ord. Dwg. No.	Stock No.
	Tester, Dynamic, M12 (including the following disassembled items:)	F8212500	J005-8212500
1	Cable Assy. Plug & Receptacle	C8212755	J675-8212755
1	Cable Assy. Plug & Receptacle	C8212751	J675-8212751
1	Cable Assy. Plug & Receptacle	C8212753	J675-8212753

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	Nomenclature	Ord. Dwg. No.	Stock No.
2	Cable Assy. Plug & Receptacle	C8212748	J675-8212748
1	Cable Assy.	C8212750	J675-8212750
1	Cable Assy.	C8212759	J675-8282759
1	Cable Assy.	C8212760	J675-8212760
1	Cable Assy.	C8212761	J675-8212761
4	Lamp	B7639999	F273-7639999
2	Fuze, 10 amp. (500546)	DGAX2	
8	Fuze, 3 amp. (106652)	DGAX1	H004-0161740

6.3 Weight and Displacement - The approximate weight and displacement of a completely packaged unit is:

Gross Weight	592 pounds
Displacement	24 cubic feet

6.4 Figures 15 thru 19 are included for information purposes only and do not necessarily reflect the specified sequence of operations.

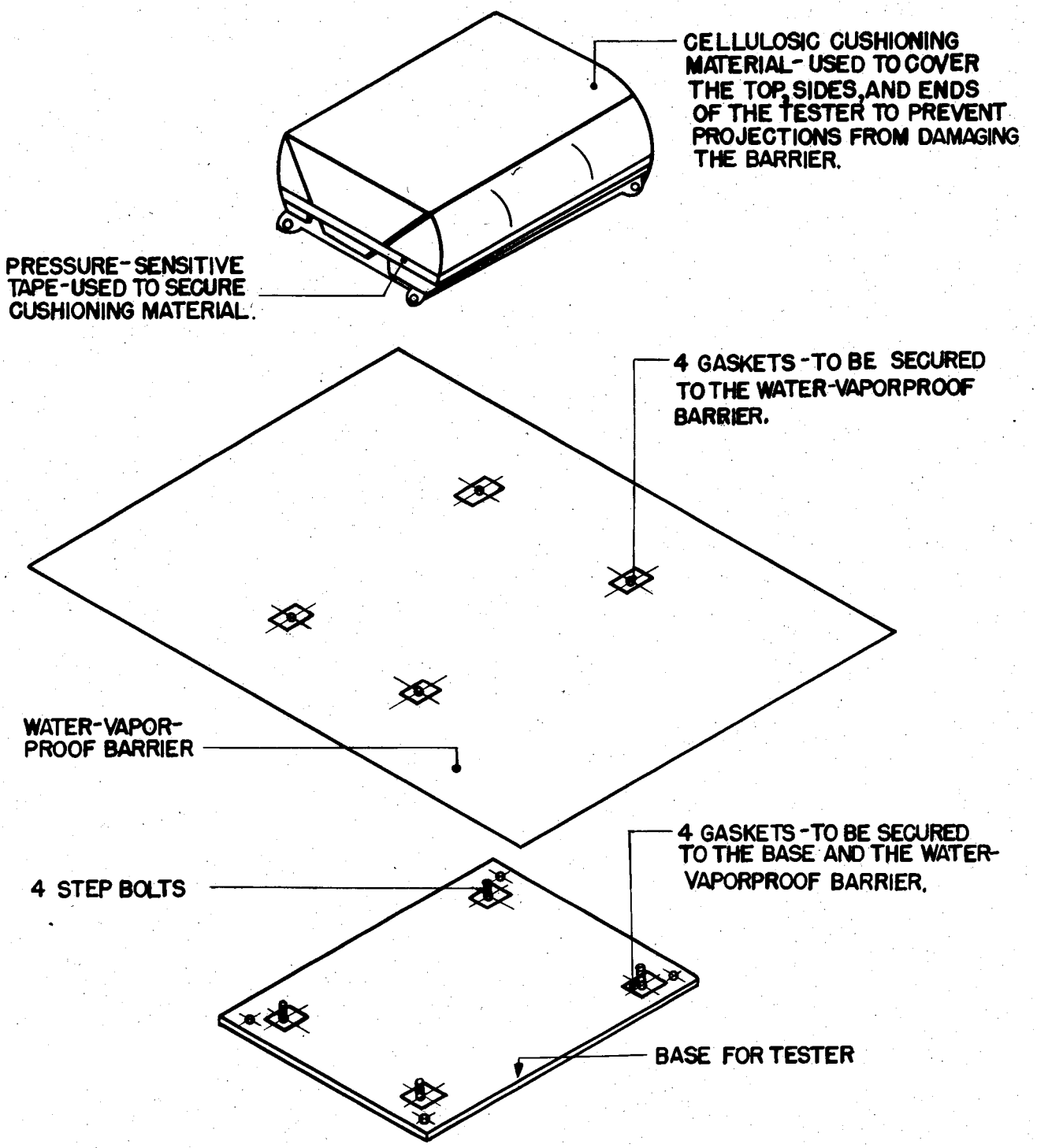
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Custodian:

Army- Ordnance Corps

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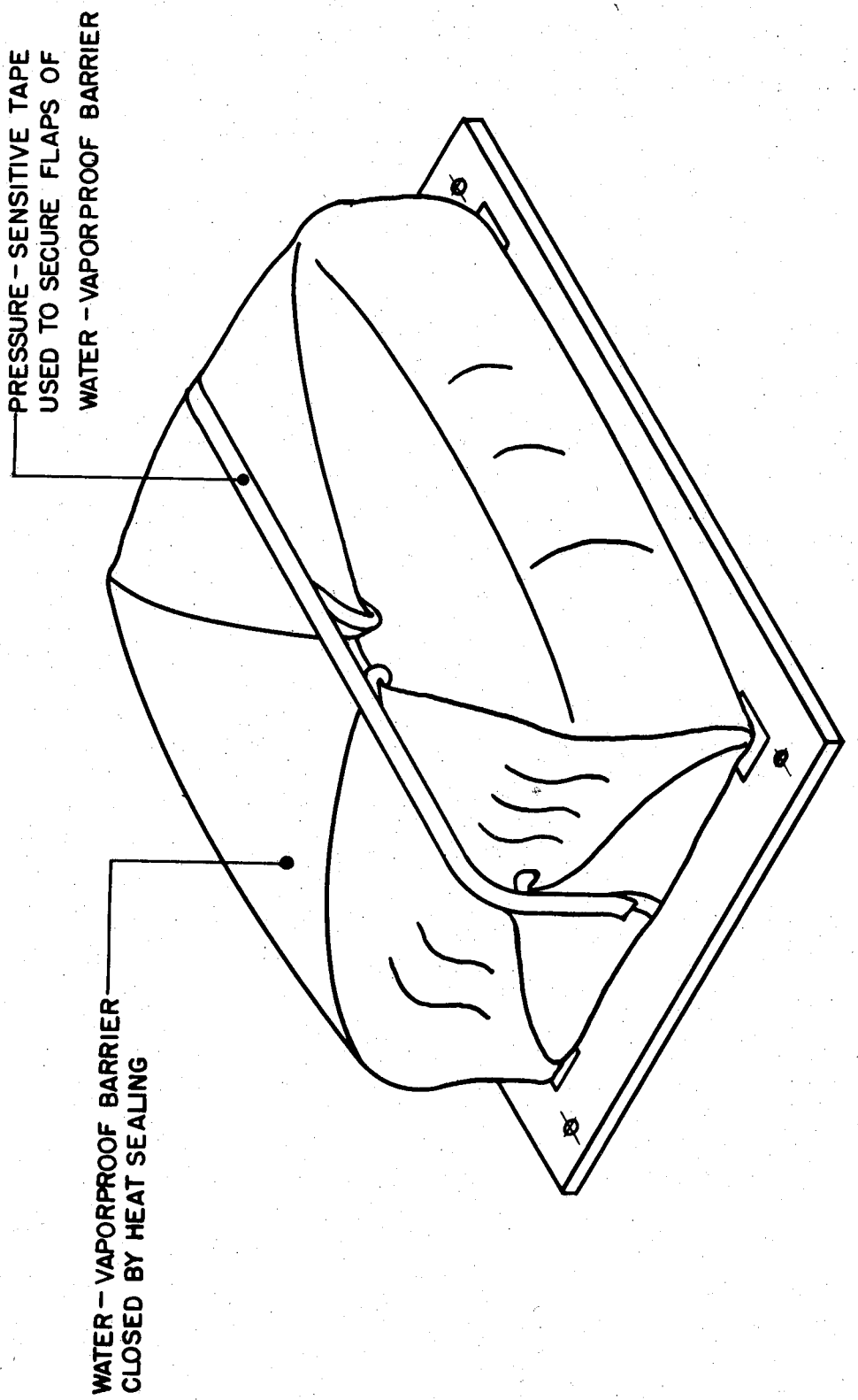
NOTE: THE HUMIDITY INDICATOR, DESICCANT, AND THE DISASSEMBLED ITEMS OF THE TESTER WHICH ARE SECURED TO THE TESTER ARE NOT SHOWN.



METHOD II a PACKAGING OF DYNAMIC TESTER

FIGURE - I

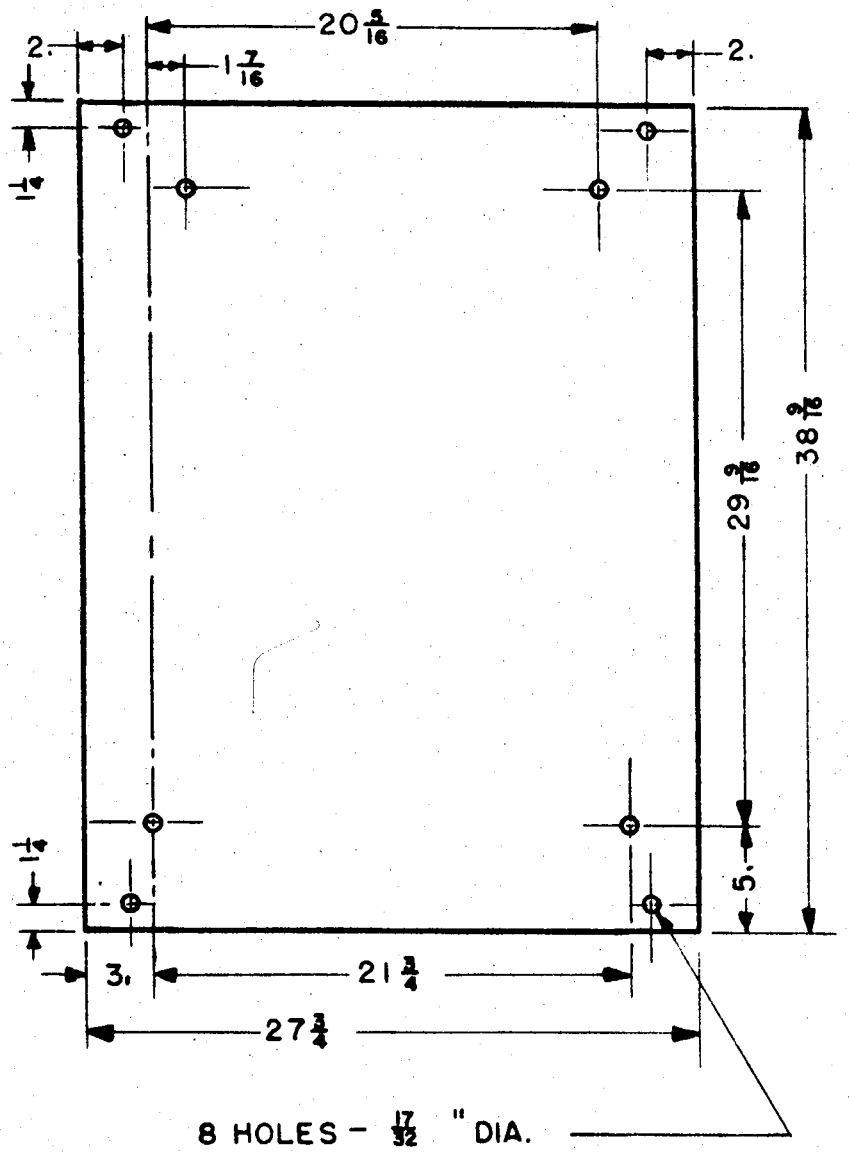
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METHOD IIa PACKAGING OF DYNAMIC TESTER

FIGURE-2

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MATERIAL:

PLYWOOD - 3/4 INCH THICK AND CONFORMING TO TYPE I OR II CLASS I OF SPEC NN-P-515

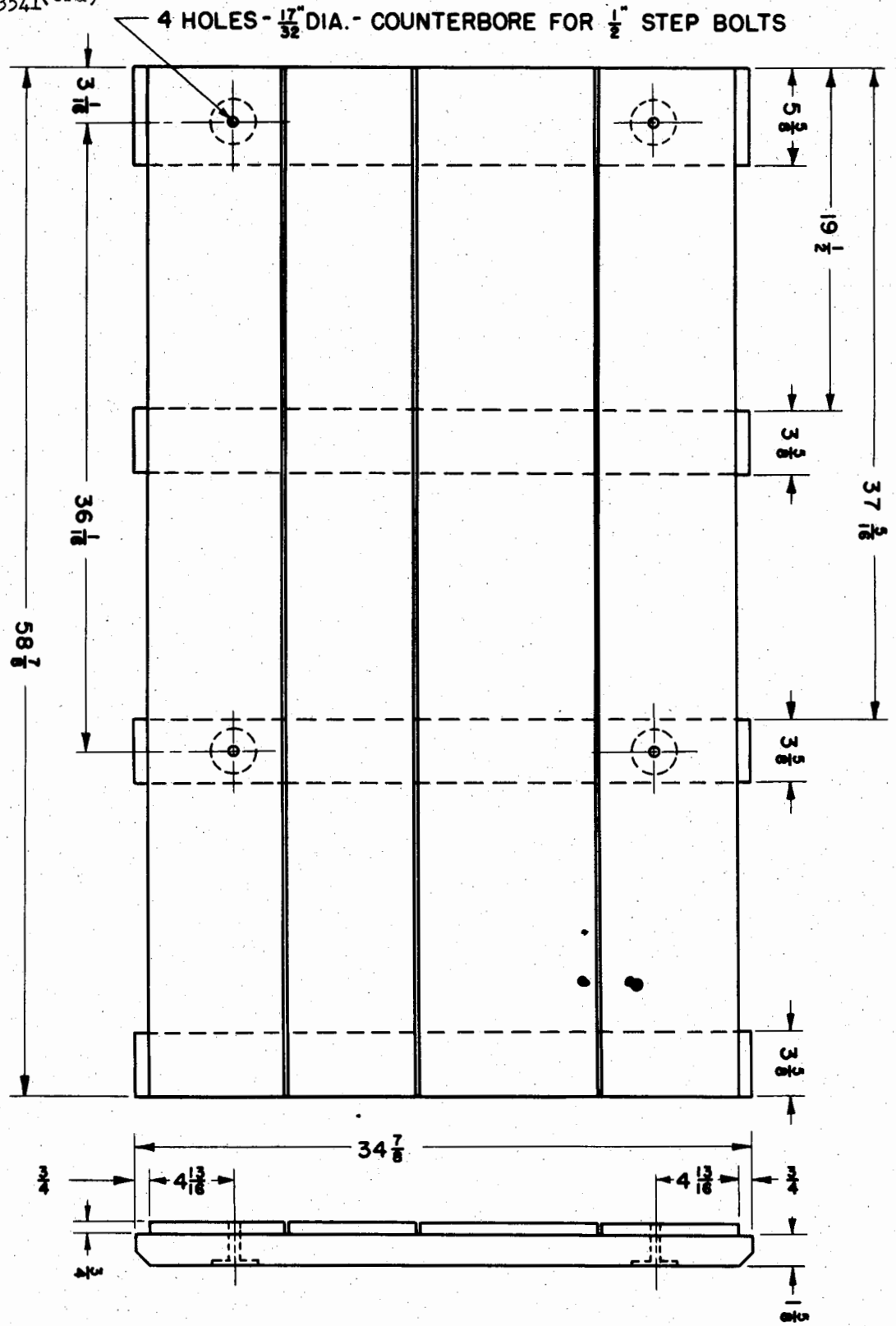
BASE FOR TESTER, DYNAMIC, M12

FIGURE - 3

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NOTES

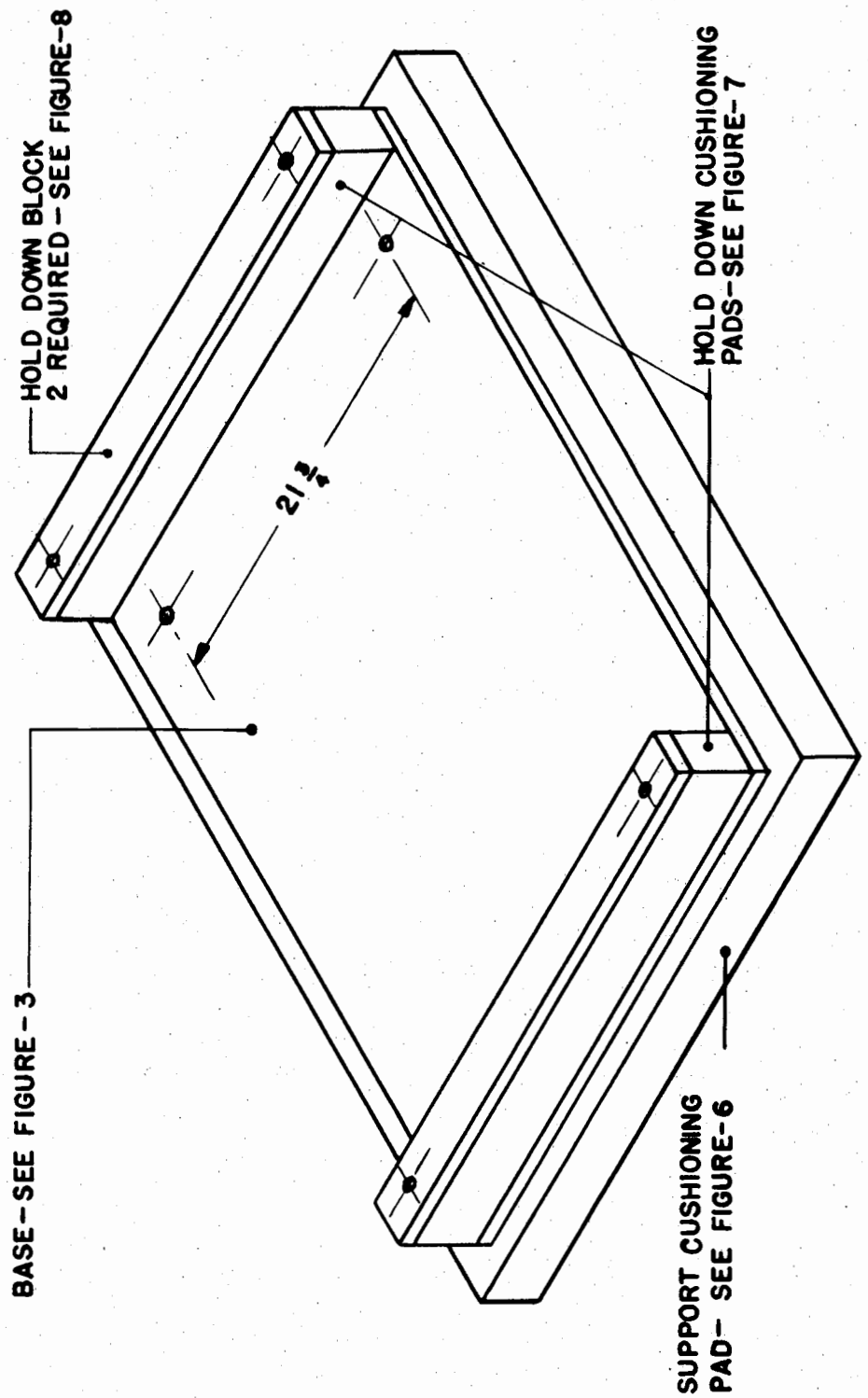
1. FLOORBOARDS SHALL BE SPACED $\frac{1}{4}$ INCH APART FOR DRAINAGE.
2. LUMBER AND NAILING REQUIREMENTS SHALL CONFORM TO SPEC. JAN-P-106.



SHIPPING CONTAINER BASE

FIGURE - 4

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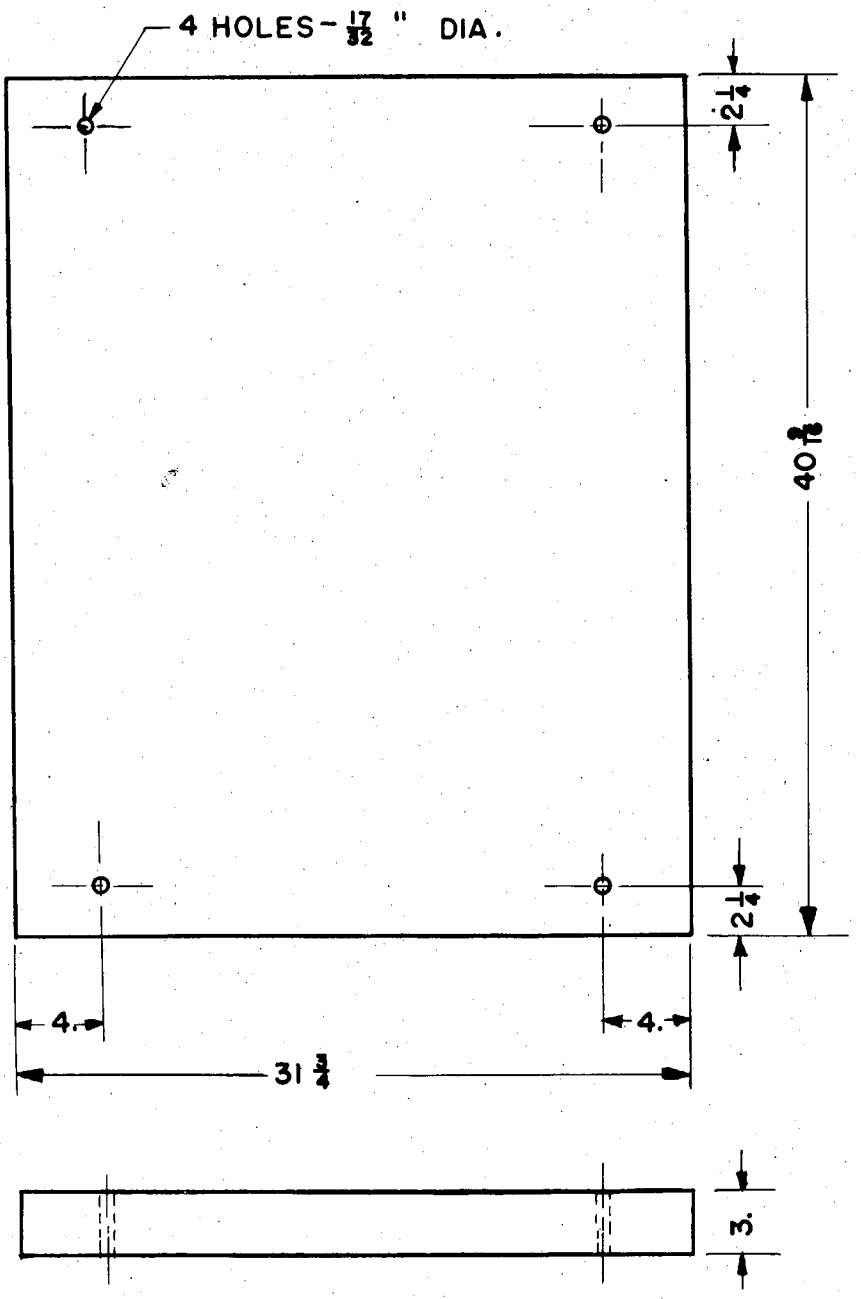
CUSHIONING PLYWOOD BASE

FIGURE-5

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MATERIAL

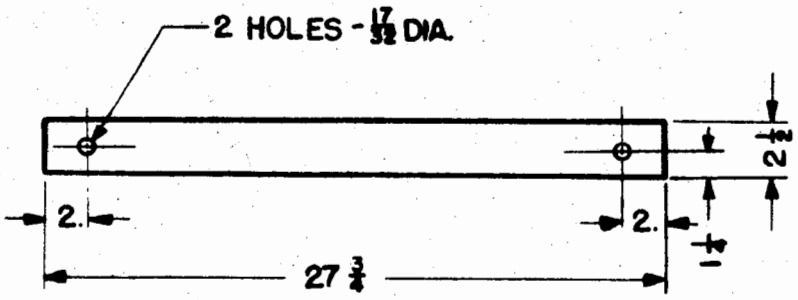
FIBROUS GLASS - DENSITY 4 LBS. PER CU. FT. AND CONFORMING TO TYPE I OF SPEC. MIL-C-17435.



SUPPORT CUSHIONING PAD

FIGURE - 6

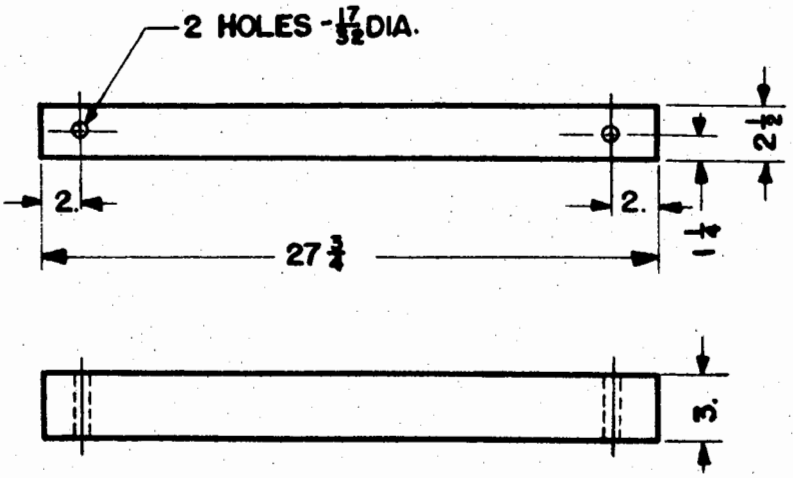
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MATERIAL:

PLYWOOD 3/4 INCH THICK AND CONFORMING TO TYPE I OR II, CLASS I OF SPEC NN-P-515

HOLD DOWN BLOCK
FIGURE - 8

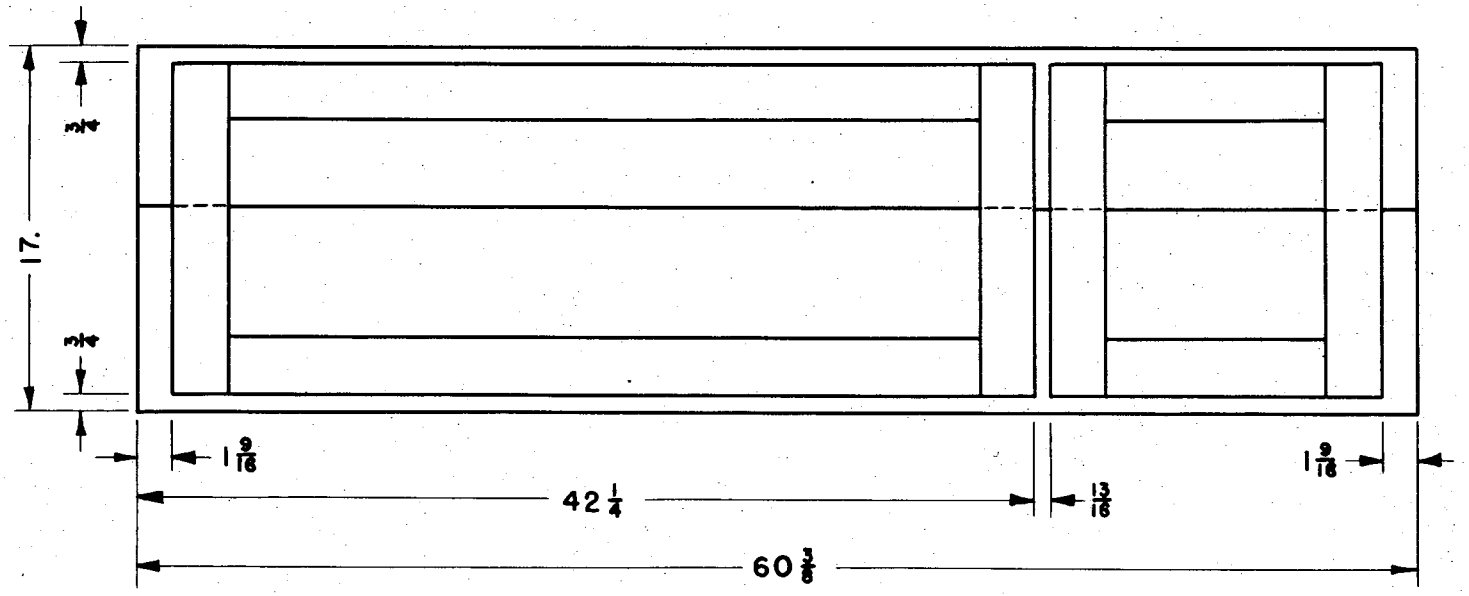


MATERIAL:

FIBROUS GLASS - DENSITY 10 LBS. PER CU. FT. AND CONFORMING TO TYPE I OF SPEC. MIL-C-17435

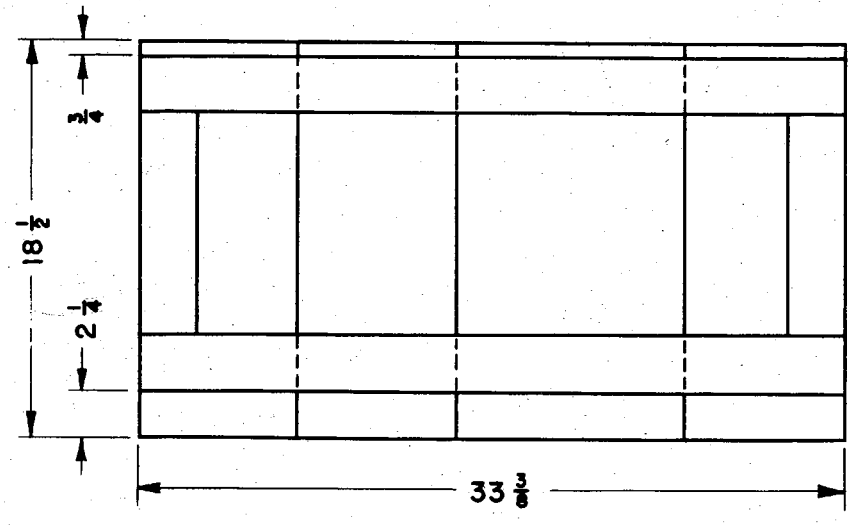
HOLD DOWN CUSHIONING PAD
FIGURE - 7

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SHIPPING CONTAINER ENDS

FIGURE - 9



NOTES FOR FIGURES 9 AND 10

1. ALL SHEATHING SHALL BE NOMINAL 1 INCH LUMBER
2. ALL CLEATS SHALL BE NOMINAL 1 X 3 INCH LUMBER
3. LUMBER AND NAILING REQUIREMENTS SHALL CONFORM TO SPEC. JAN-P-106.

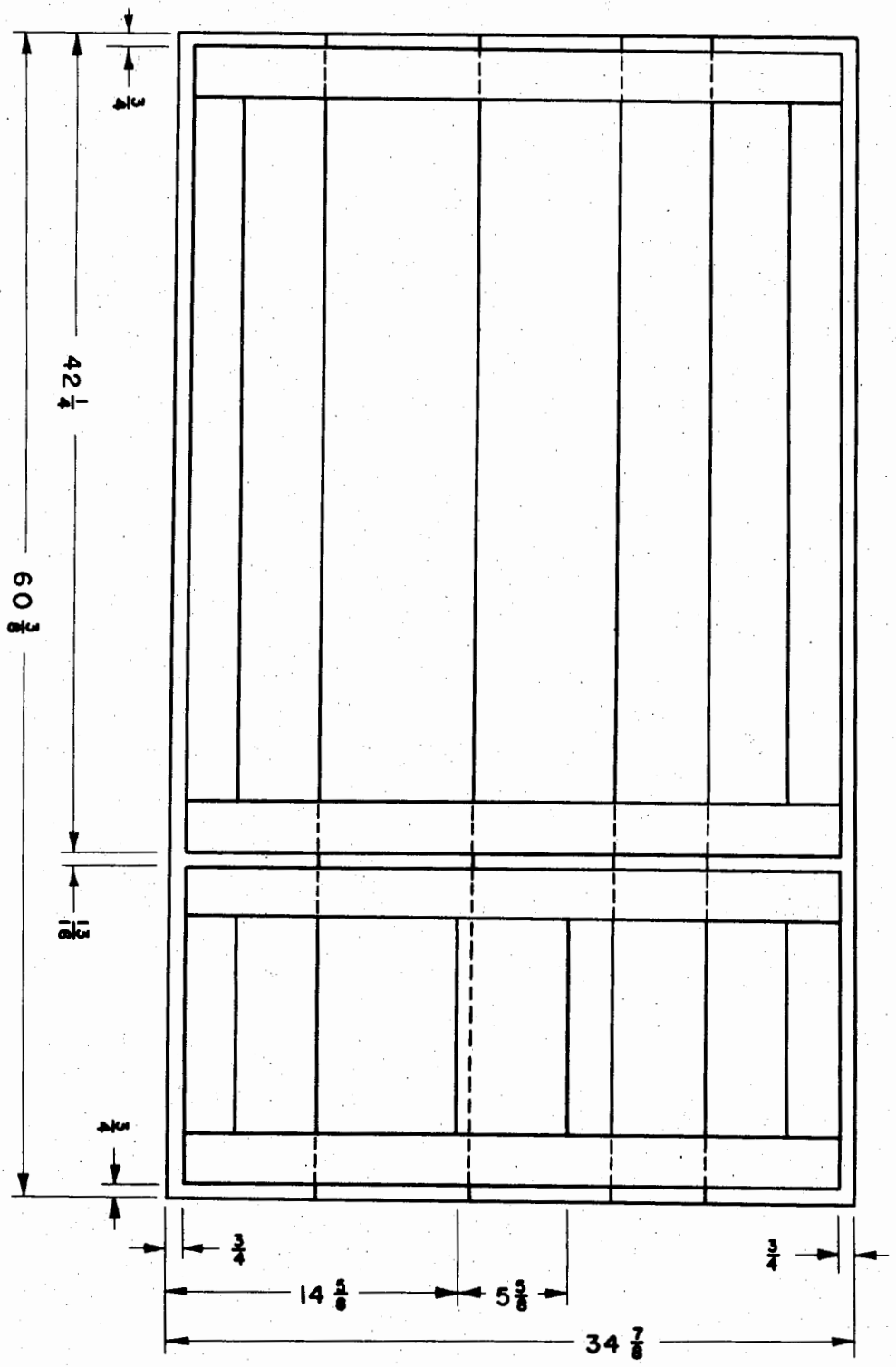
SHIPPING CONTAINER SIDES

FIGURE - 10

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NOTES

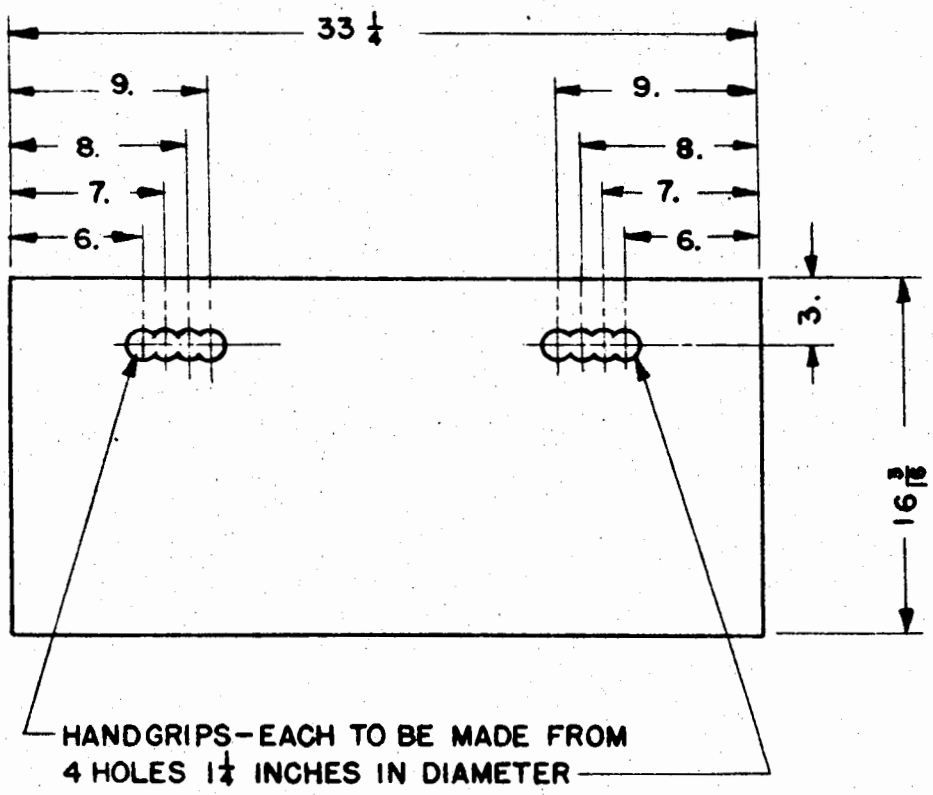
1. ALL SHEATHING SHALL BE NOMINAL 1 INCH LUMBER.
2. EXCEPT AS INDICATED ALL CLEATS SHALL BE NOMINAL 1X3 INCH LUMBER.
3. LUMBER AND NAILING REQUIREMENTS SHALL CONFORM TO SPEC. JAN-P-106.



SHIPPING CONTAINER TOP

FIGURE - 11

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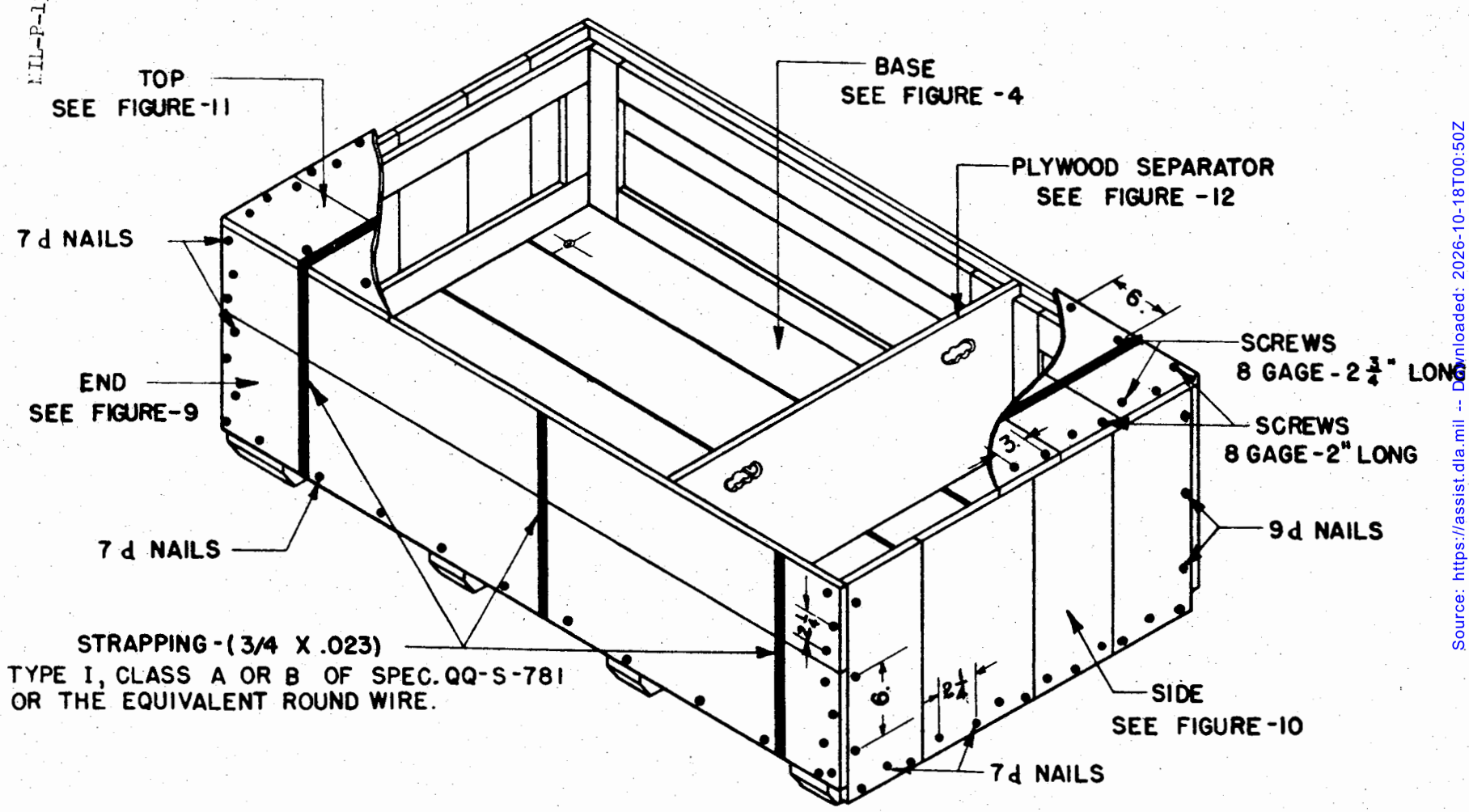
MATERIAL:
PLYWOOD-3/4 INCH THICK AND CONFORMING TO TYPE I OR II, CLASS I OF SPEC NN-P-515

PLYWOOD SEPARTOR

FIGURE-12

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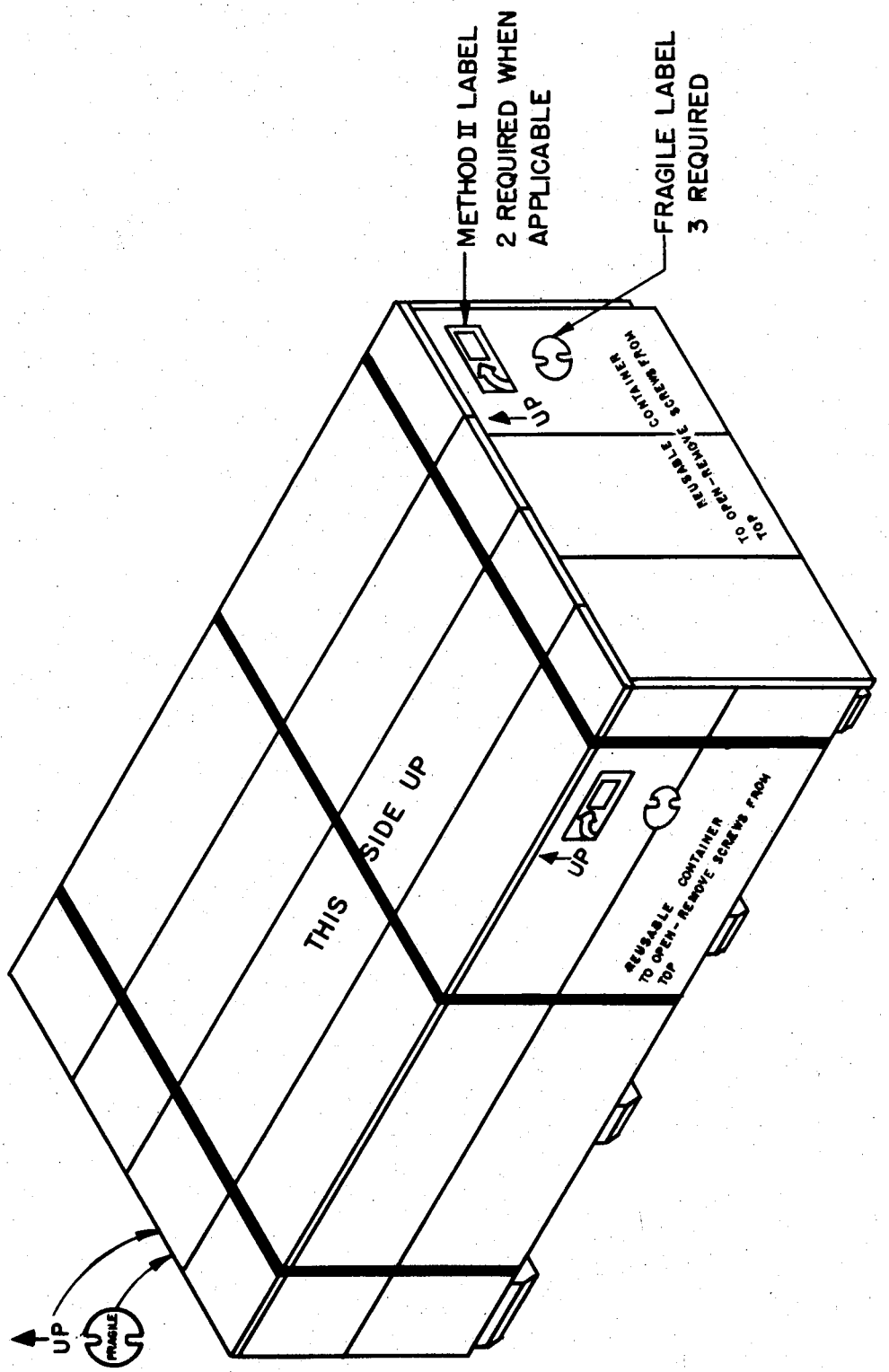
NOTES: 1. NAILS SHALL BE CEMENT COATED AND CONFORM TO SPEC. FF-N-105
2. SCREWS SHALL CONFORM TO SPEC. FF-S-111.
3. THE CUSHIONED TESTER AND THE PACKAGED DISASSEMBLED ITEMS ARE NOT SHOWN.



ASSEMBLING AND STRAPPING OF SHIPPING CONTAINER

FIGURE - 13

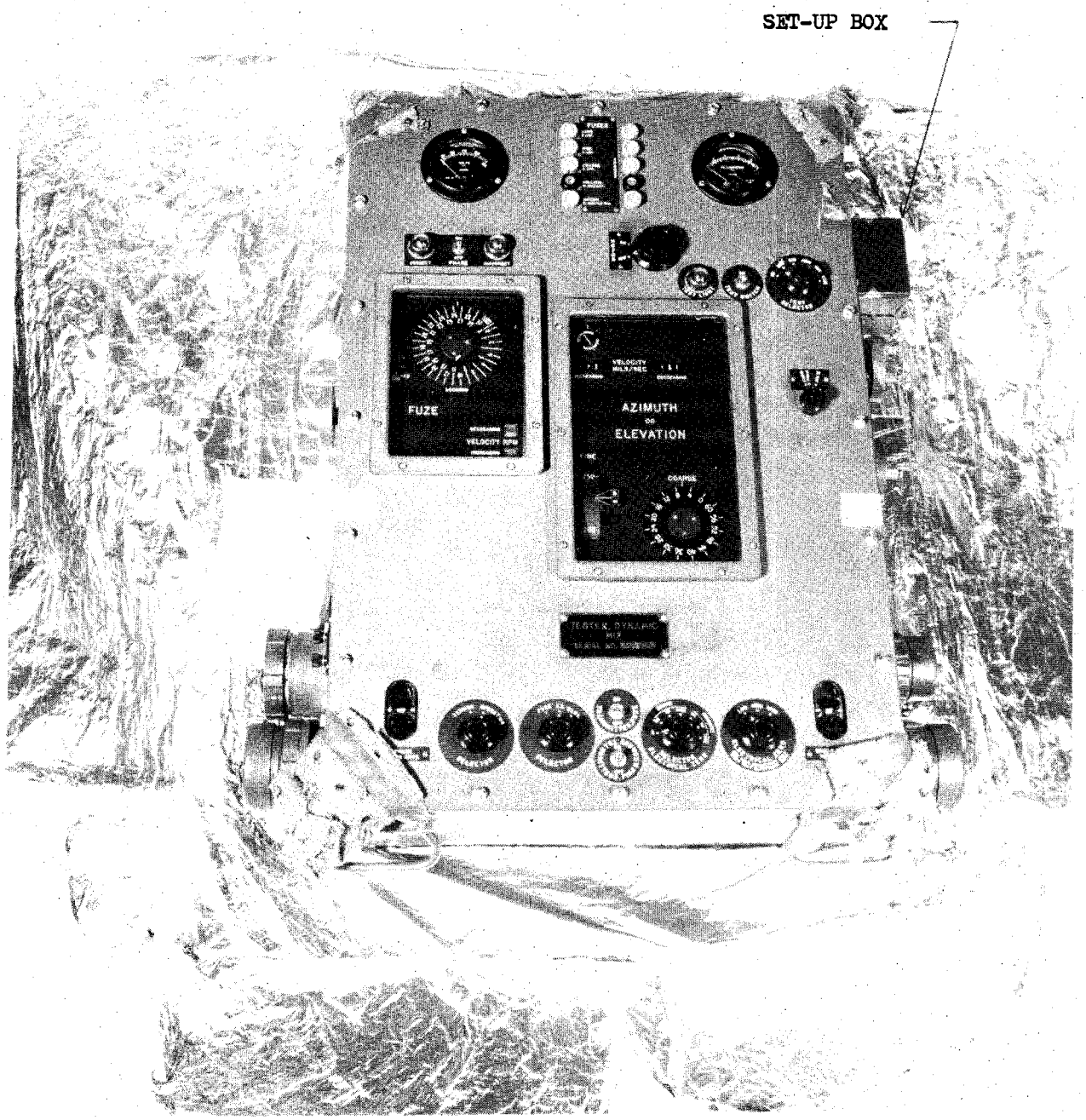
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SPECIAL HANDLING AND PRESERVATION MARKINGS

FIGURE-14

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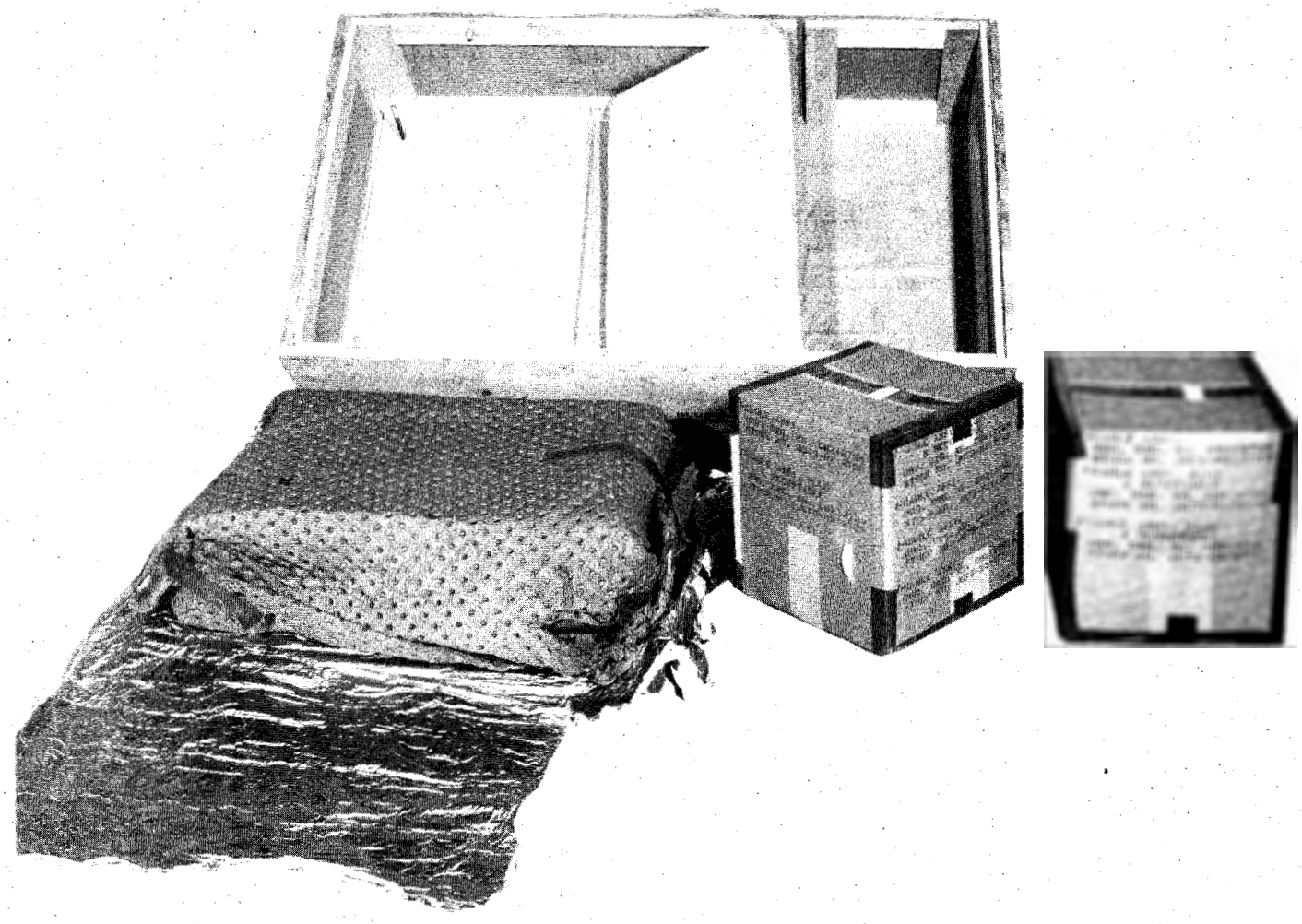


NEG. #23947-1 7-28-53 FRANKFORD ARSENAL ORD. CORPS

TESTER, DYNAMIC, M12 - Tester on barrier material, showing cushioning material on corners, desiccant secured, set-up box containing lamps and fuzes.

FIGURE 15

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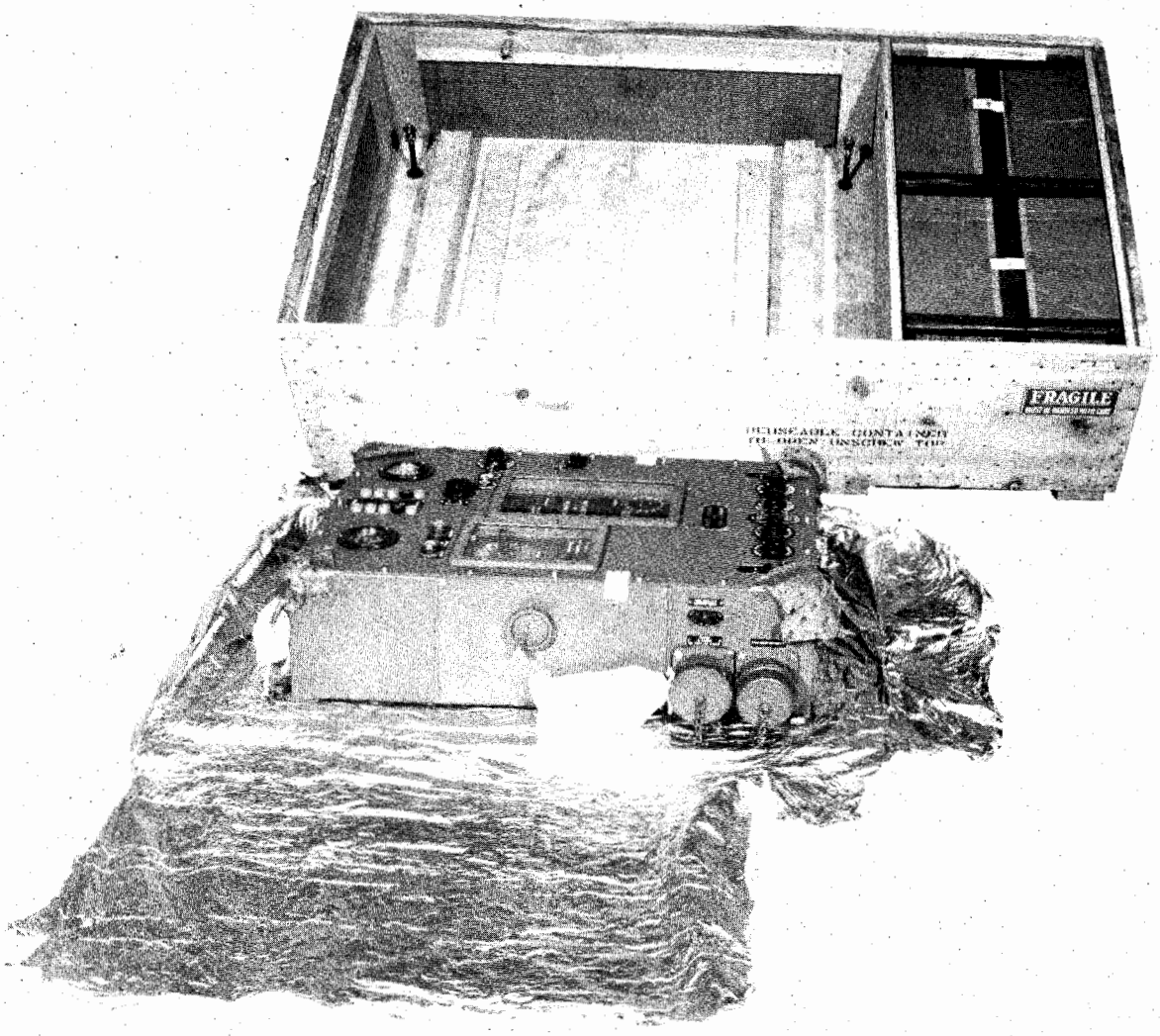


NEG. #23947-2 7-28-53 FRANKFORD ARSENAL ORD. CORPS

TESTER, DYNAMIC, M12 - Tester cushioned to prevent rupture of barrier material, shipping container with support pads, cartons containing cable assemblies.

FIGURE 16

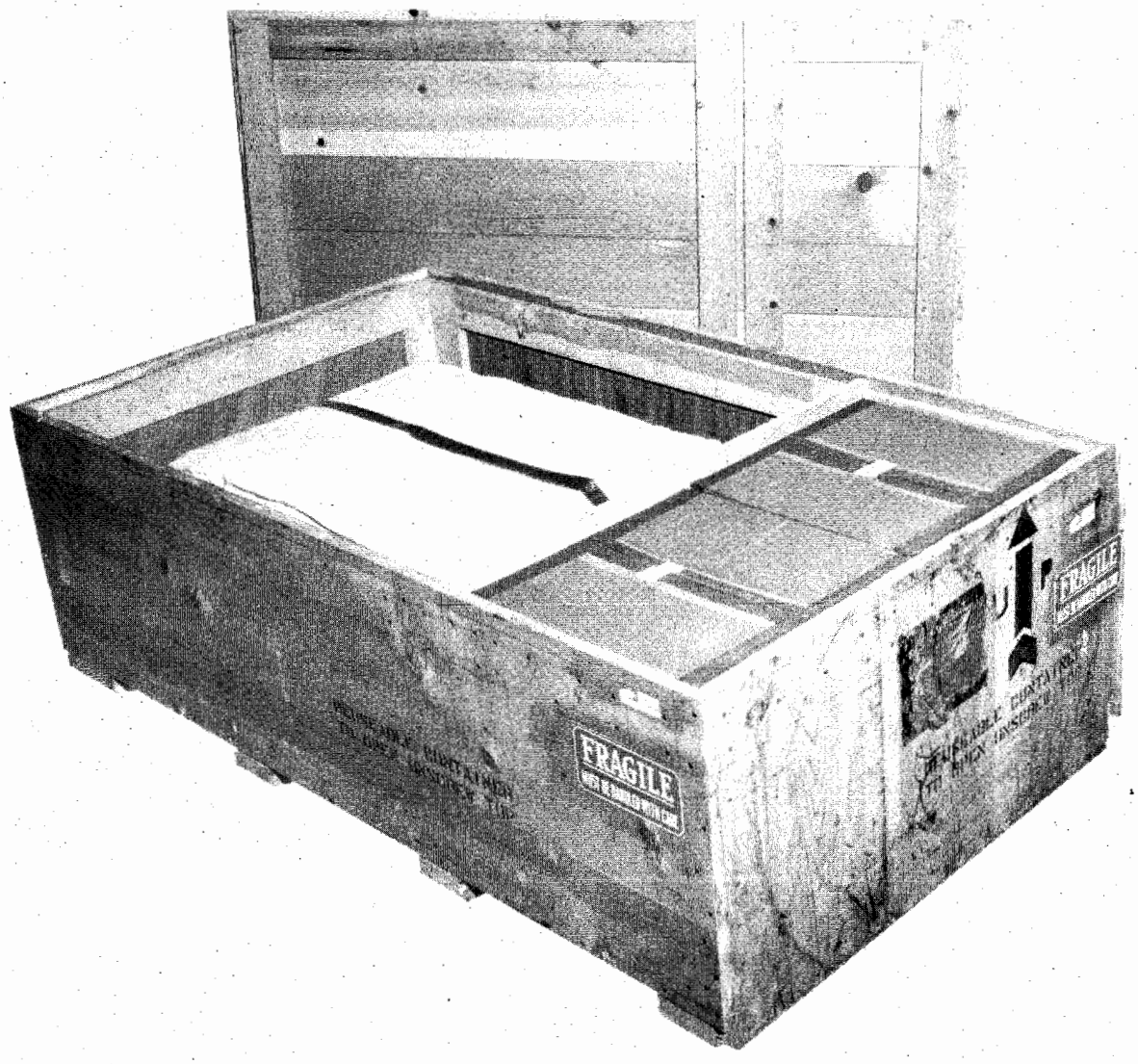
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NEG. #23947-3 7-28-53 FRANKFORD ARSENAL ORD. CORPS

TESTER, DYNAMIC, M12 - Plywood separator and cartons containing cable assemblies positioned in shipping container.
FIGURE 17

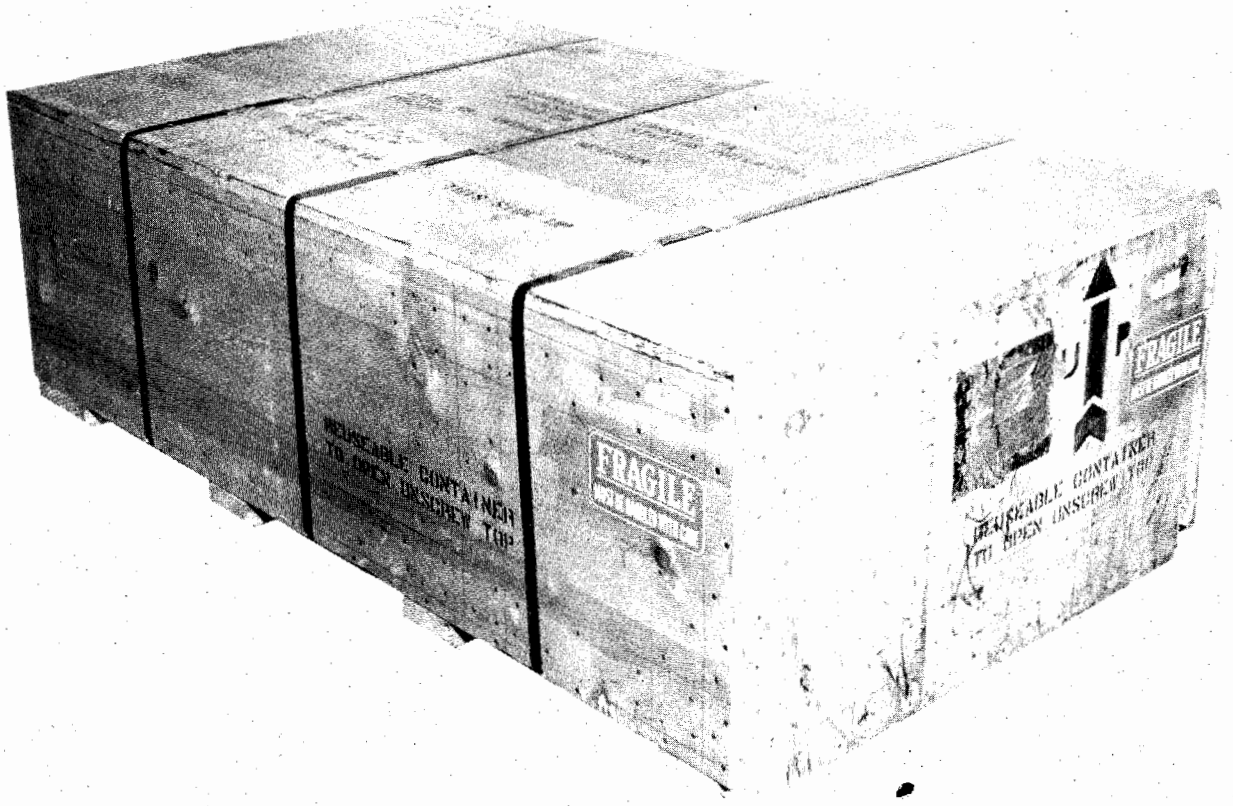
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:50Z
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NEG. #23947-4 7-28-53 FRANKFORD ARSENAL ORD. CORPS

TESTER, DYNAMIC, M12 - Completed pack prior to securing cover.
FIGURE 18

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NEG. #23947-5 7-28-53 FRANKFORD ARSENAL ORD. CORPS

TESTER, DYNAMIC, M12 - Completed shipping container, less identification marking.
FIGURE 19

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